

Initiating Interactions and Negotiating Approach: A Robotic Trash Can in the Field

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Abstract

In this study, we address how people respond to a robotic trashcan initiating interactions and offering its service. We show that considerable coordination and negotiation work takes place both between human and robot and between humans who are involved in a joint activity when the robot approaches. While in this scenario attention getting was no problem, the interactions posed significant problems to people who did not want the robot's service. The unwillingness to interact with the robot was mostly communicated by withholding social signals, which means that human-robot interaction designers not only need to build in ways to respond to human social signals in a timely and appropriate manner, but also a representation of what kinds of signals could be expected in order to interpret the ostensive lack of such signals adequately.

Introduction

Initiating interaction involves a series of small-scale intrusions: a) getting a person's attention—which may interrupt an ongoing activity, b) assessing that person's willingness to interact—which may violate the person's privacy, and c) creating physical proximity to enable interaction—which may violate the addressed person's personal space (Hall 1969). Since these actions are all potentially problematic, they need to be negotiated between the person (or robot) approaching and the person or people being approached. Such negotiations between people commonly involve the exchange of subtle social signals and are often carried out implicitly rather than explicitly because possible face threats are involved (cf. Brown & Levinson 1987). That is, imposing one's presence onto another person by getting his or her attention threatens that person's need for freedom from imposition. Likewise,

coming close enough for the interaction to happen may threaten the need for freedom from imposition of the person approached. Conversely, telling a person that one is not willing to interact with him or her threatens that person's need for positive face, i.e. the wish to be accepted as one is (Brown & Levinson 1987). The initiation of interaction is thus a highly subtle, coordinated activity (see also Schefflen 1972, Kendon 1990, Clark 1996).

What, then, are the social signals by means of which people indicate to an approaching robot whether they want to interact with it or not? In the current study, we investigate a) how people respond to the robot's approach in terms of attention, b) how people signal to the robot that they want to interact with it, and, alternatively, c) how people try to let the robot know that they don't want to interact with it. The signals identified will need to be taken into account to provide robots with socially acceptable methods for coordinating with people.

Previous Work

Previous studies on initiating interactions with people unfamiliar with the respective robot have shown that getting people's attention is not trivial (see Satake et al. 2009, Fischer et al. 2014ab), and getting them to interact with the robot or even to help it can be very challenging (Hüttenrauch & Severinson-Eklundh 2003, Fischer et al. 2014a). In the study by Hüttenrauch & Severinson Eklundh (2003), the robot tried to ask people in an office for help. The authors show that whether people are engaged in a task has a considerable impact on their willingness to assist the robot, whereas previous information about the robot, its capabilities and its task had only little effect.

Furthermore, Sidner and colleagues (2005, 2007) address how a robot can initiate interactions with people in an

office setting; they distinguish between situations in which the robot perceives a person but needs to get the person's attention and situations in which the robot is not even certain that someone is around. In situations in which a participant is perceived to be attending, the robot looks at and greets the person. In contrast, if a face is detected as not turned towards the robot, the robot produces a verbal request. If the robot expects someone to be around but cannot identify anyone specifically, it just produces a question and waits for someone to self-select for a response. The principle under which the robot is operating is to use as many reliable sources of knowledge as possible to make an informed decision, and otherwise to be minimally disturbing. However, the authors also report that the principle of minimal disturbance may not always be effective and that getting people's attention can be quite challenging (Sidner et al. 2005).

In a study by Fischer et al (2014a) in which participants were busy filling out forms and were not aware that they were participating in a study already, none of the participants responded to the robot beeping to get attention, and very few were willing to interact with the robot in order to assist it. In particular, we found that busy people may react only reluctantly to the robot's gesture and generally do not respond to a robot beeping at all; in contrast, all participants reacted to the robot's use of speech by looking at the robot. However, people were still unwilling to interrupt their own activities, and few people stepped up to help the robot.

Related to the finding regarding the relative success of speech to get a person's attention, Tellex et al. (2014) find that having the robot make specific requests for assistance based on a common understanding of what the person can perceive significantly improved people's willingness and satisfaction in assisting a robot, compared to a condition in which more generic requests for assistance were made.

Concerning the effectiveness of acoustic signals, Fischer et al. (2014b) investigated the effectiveness of different beep contours and confirmed that beeps are not very well suited to get people's attention if they are busy or engaged in a social encounter, but that the melody of the beep sequence nevertheless has an effect on the perceived friendliness of the robot; thus, it is not the case that people would not hear the beep.

Similar problems were encountered by Satake et al. (2009), who tried to have a robot engage with people in a shopping mall. Many people were not willing to interact with the robot, and many others did not notice the robot or understood its intentions. Predicting people's moving trajectories improved getting people's attention and making them understand that the robot wanted to interact, yet still many people refused to interact with the robot.

To sum up, these studies suggest that getting a person's attention may be quite challenging for a robot, especially if the person is busy.

Relevant previous work furthermore comprises research on comfortable distance and direction of approach; one such study is Walters et al. (2007), which addresses the influence of the direction from which the robot should approach. The authors find that people do not like the robot to approach from the back. Similarly, they find that people's degree of comfort when the robot approaches also depends on whether they are sitting or standing. Concerning distance, Yasumotu et al. (2011) investigate people's preferences regarding the distance to the humanoid robot Asimo. Participants were asked to indicate when the robot should not come closer, which on the average was at about 78cm.

Mumm and Mutlu (2011) investigate the effects of likeability of the robot (manipulated by means of polite versus rude introductory statements by the robot) and eye gaze (mutual versus averted) on the distance that participants keep when they move around the robot themselves. The authors find that in the mutual gaze

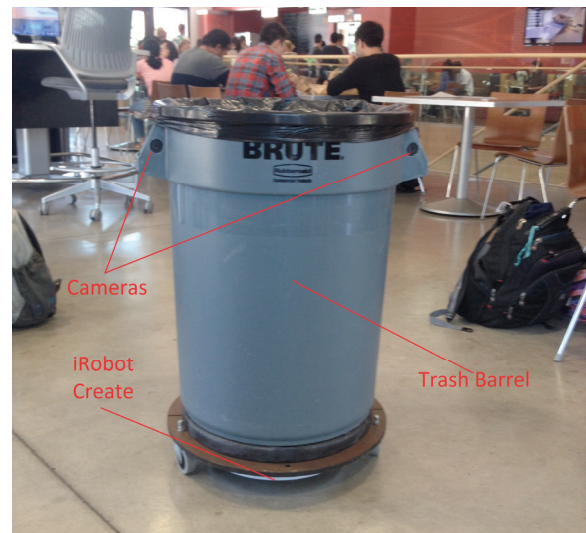


Figure 1: Trashcan robot

condition, participants increased the distance to the robot; similarly, if participants disliked the robot, they also increased the physical distance. However, if participants liked the robot, gaze had no influence on physical distance. These results correspond to earlier findings by Takayama and Pantofaru (2009) who report negative attitudes towards robots as well as the personality trait of neuroticism to be the best predictors of the distance people place themselves in when interacting with a robot. Furthermore, previous experience with robots and pet-ownership make people decrease the distance to a robot.

These studies show that the distance people prefer the robot to keep depends on various factors, which makes the

physical approach of a robot potentially problematic. In interactions between humans, people have been found to negotiate subtly and implicitly physical distance and positioning in space in relation to each other (see Kendon 1990, Schefflen 1972, Schegloff 1998). We thus assume that also distance in HRI needs to be negotiated to be coordinated between human and robot.

Experimental Study

The study carried out is a naturalistic field trial of a robotic trashcan. Participants were engaged in the quasi-public setting of a university dining area, and were not forewarned that they might be engaged in a study. The robot was remotely operated.

Robot

The robot is a trashcan mounted on a base driven by a iRobot Create and equipped with two cameras hidden in the handles of the trashcan. The robot was remotely controlled by a wizard who was seated in the cafeteria and who only had the video footage and audio from the robot's embedded cameras available to steer the robot. The robot was wirelessly operated via a laptop, so that from the outside it was not apparent that the operator was steering the robot, and in fact none of the people in the cafeteria succeeded in identifying the operator, even though many looked for him (or her).

The robot's behavior was scripted: initially, the robot approached a table by driving up to it in 1.5m distance, then wiggled and waited until invited to come closer. Furthermore, if the participants were perceived to throw trash into the robot, the robot would wiggle. In the second phase of the experiment, the robot drove directly up to the table and even aimed at bumping into something, like a nearby chair, the table or another obstacle.

Experiment Site

The field trial was carried out in a public cafeteria on the Stanford campus and took about one and a half hours. The data recorded are the video recordings from the robot's inbuilt cameras. In addition, three researchers were engaged in observing and interviewing participants during this experiment.

Method and Data

The method for analysis is based on ethnomethodological conversation analysis and concentrates on the sequential turn-by-turn analysis of observable social signals in interactions (cf. Sacks 1984).

Altogether 26 interactions were recorded over the course of 1:16 hours, five of which involved single participants; in addition, other encounters were observed and will briefly be discussed as well.

Results

Observation shows that people moving—for instance, people who were just passing through the building—did not realize that the robot was more than an ordinary trashcan. These people just walked by, sometimes relieving themselves of trash, but they did not respond to the robotic aspects of the trashcan at all.

Furthermore, there are people who just watched the robot from afar and who were never involved in the interactions themselves. Likewise, many of the participants in the interactions for which video footage is available had observed the robot before it was approaching, and thus attention getting is not captured in the videos in these cases.

Getting Attention

The video analysis of how the robot gets people's attention reveals a clear difference between people who are alone at their table and people who are in the company of other people.

In company, attention to the robot is negotiated between participants because attention to the robot means withdrawing attention from the co-participants. A typical example of such a negotiation process is the following interaction. At this table, two people, one participant on the left (LP) and one on the right (RP), are seated facing each other (Fig. 2):¹

R: approaches (appr. 1.5m)

LP: turns head briefly to R

RP: follows LP's eye gaze and turns head briefly to R, then back to LP, laughs and says something

R: wiggles

RP: smiles, looks towards

LP, says something

LP: looks at R smiling and slowly lifts a package of trash and lowers it again

RP: looks towards R and back to LP, laughs and says something

R: drives closer (appr. 80cm)

LP: fixes gaze on RP, smiles only little, turns head to look around



Figure 2: Participants orient towards each other before the robot is approaching

¹ We show pictures of those participants who gave their consent in the interview after the recording; if people were not interviewed or did not give their consent, their faces in the pictures were obscured.

RP: gets up and throws trash into the robot, laughs
 LP: laughs and throws in his trash, looks at R
 R: wiggles
 RP: looks at R and in a direction between R and LP, laughs, says something
 LP: looks at RP, laughs

When the robot (R) approaches, the person on the left (LP) turns his head towards the robot, and the communication partner on the right (RP) follows his eye gaze to the robot and begins to smile when turning back to looking at LP. When the robot wiggles, both participants laugh while looking at each other. When they then interact with the robot, they still look back and forth between robot and communication partner. Thus, whether people attend to the robot and how it is to be treated is continuously negotiated with the co-participant.

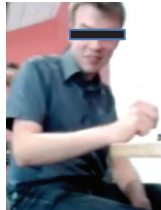


Figure 3: Male participant (M) turns to attend to the robot

An example in which this negotiation results in continuing with the current activity is the following encounter in which a female (F) and a male participant (M) are involved:
 M: sees robot and turns his head towards it, smiling (see Fig. 3)
 F: continues to speak while looking at her laptop, then looks at M
 M: says something, looking at F
 F: smiles briefly, then looks at her laptop again, starts speaking and then typing (see Fig. 4)

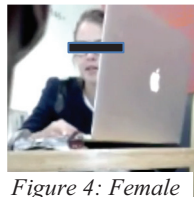


Figure 4: Female participant (F) keeps working on her laptop

To conclude, people in company need to ratify devoting attention to the robot with their communication partners. Only if all partners at the table agree to attend to the robot, people will interact with the robot.

Signaling Willingness to Interact

Almost all groups of people the robot encountered during the field trial jointly decided to interact with the robot. They displayed this willingness to interact by waving at the robot, waving with trash, calling the robot, getting up, changing their body orientation etc. They all smiled and laughed (cf. Kirstein et al. 2015). An example of such

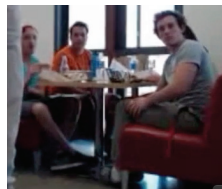


Figure 5: Participant on the front right (FRP) turns to see the robot

actions is given by a group of four participants²:

FLP, BLP and FRP are observing the robot from afar. FRP rotates his body to see better (Fig. 5).

FLP says something to BLP while both continue looking at R.

R approaches, FRP turns back to the table and picks up some trash, saying something.

FLP and BLP laugh

BRP turns her body around to throw trash into the robot from behind FRP.

FLP and BLP smile and FLP waves at R to come closer (Fig. 6).

R comes closer

FLP waves smiling at R to come closer

R comes closer

FLP and BLP lean forward to throw trash into R.

FRP says something about the robot having sound sensors
 FLP touches the garbage bag in the robot, saying that he wants to make sure that he is not covering any cameras

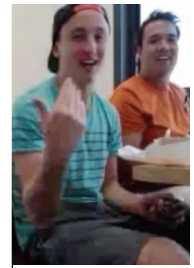


Figure 6: Participant on the front left (FLP) waves at the robot

Thus, these participants use body torque, eye gaze, waving, gesturing and verbal utterances to interact with the robot.

Another example is from a group of four women who were already sitting tight with their trash ready when the robot approached:

BRP and FRP are watching R, laughing, talking to each other while looking at R. FRP holds her trash in hands and raises her arms when the robot approaches

R: approaches (appr. 1.5m)

FRP throws in her garbage (Fig. 7)

BRP shouts ‘ahhhh’ in approval and throws in her garbage while R is coming closer

FRP picks up further garbage from the table and drops it into the robot

BRP is laughing and cheering, also the participants on the left side of the table can be heard laughing

FRP says “thank you thank you thank you!”

The women get up laughing and leave

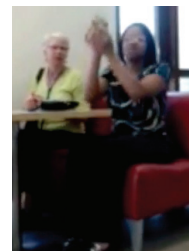


Figure 7: Participant on the front right (FRP) prepares to throw trash into the robot

Single participants who wanted to interact with the robot displayed their willingness by holding out trash, which is however generally done more subtly than by the groups of people, as the following example illustrates:

² Participants are named with respect to their position around the table, i.e. front left (FLP) and back left (BLP) as well as front right (FRP) and back right (BRP).

The person notices the robot from afar, empties his bottle, changes his seating in such a way to make room for the robot to approach (by pulling his legs under his body and by sitting up straight), but the robot does not come close enough. He shakes and wiggles the bottle. The robot wiggles, but to get the attention from the couple nearby – the operator has in fact not noticed the signals produced by this person. The person then gets up and throws his bottle into the robot on the way out.

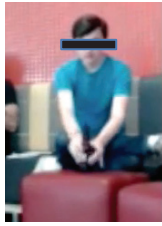


Figure 8: Participant wiggles his empty bottle at the robot

Displaying no Willingness to Interact

If people did not want to interact with the robot, different kinds of behaviors were observed; for instance, one participant kicked the robot to make it go away:

R: approaches (appr. 30cm)

P: briefly looks up from book at R then down again, then glances at R again, unsmiling, and back to the book

R: wiggles

P: glances at R again, unsmiling, then back at the book

R: backs up a bit, wiggles again

P: looks at robot, looks back at book

R: wiggles again

P: looks at robot, leans back, moves leg up and kicks R while continuously looking at R. Then suddenly sits up again while still looking at R before sitting back and looking at the book again

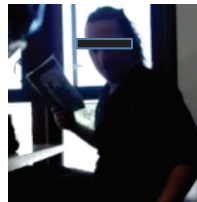


Figure 9: Participant sits up after kicking the robot

In the interview, this participant said that he did not want an unclean object like a trashcan too close, and that since he didn't have any trash, he wanted the robot to leave right away (note that the whole episode lasted 26 seconds).

However, much more common was demonstrated withholding of social signals, i.e. withholding of eye gaze and averted body orientation. In the following example, the participant uses both averted eye gaze and body posture as well as speech to make the robot leave:

P: intensely typing

R: approaches (appr. 1.5m)

P: looks up at R, continues typing

R: comes closer (appr. 80cm)

P: looks at R, smiling

R: wiggles

P: keeps looking at R, then back at laptop, then back at R smiling and says: "no trash" very slowly

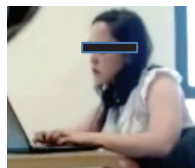


Figure 10: Busy participant retains her body orientation

R: moves away

P: watches the robot move away, then returns to laptop, still smiling

The next example describes an interaction between a person who is busy eating and typing into her mobile when the robot arrives:

R: approaches (1.5m)

P: briefly glances at robot (without moving her head) and back at mobile

R: wiggles

P: glances at robot and back at mobile

R: wiggles

P: glances at R, looks at mobile and continues eating

R: wiggles and comes closer (1m distance)

P: gazes at robot while it is moving without moving her head, then looks in opposite direction, continues eating, tilts head in opposite direction

R: moves closer, pushes chair aside, until it reaches the table, wiggles

P: briefly glances at robot, then looks straight ahead and down at phone, tries to call someone, looks around in opposite direction without moving her head.

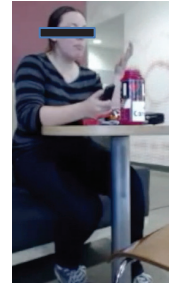


Figure 11: Participant withholding any social signals

Thus, the only way in which this participant signals to the robot that she does not want any interaction is by not producing any social signals; even when she looks at the robot, she does not move her head in order not to display any orientation towards the robot.

In more cases than not, participants who withheld social engagement cues when the robot approached, did glance back or follow the robot with their gaze as it departed, suggesting that their response was not just due to lack of interest or awareness.

Another case is interactions in which people willingly interacted with the robot but had no trash to fulfill the functional purpose of the interaction. In this case, people produced trash – one group tear off the lower part of their order receipt to 'feed' it to the robot. In the interview they said that they felt that the robot 'wanted' trash.

Discussion

Initiating interactions by the trashcan robot has turned out to be highly contingent on the kinds of activities people are engaged in. In particular, people engaged in social

encounters negotiated with their communication partners whether an interruption is acceptable; in case the interaction partners ratified the distraction, people willingly engaged in interaction with the robot. If however one of the partners insisted on continuing with the ongoing activity, people did not engage with the robot. Thus, we can conclude that people do not easily interrupt their activities; if they are doing something else that is important, they will ignore the robot. This ties in with results by Hüttenrauch & Serinson-Eklundh (2003) and by Fischer et al. (2014a), who also found that people do not attend to robots when they are engaged in their own activities.

Likewise, people who are engaged in activities on their own may interrupt their activities to interact with the robot – however, in our data they only did so if they had trash and thus wanted the robot's functionality. None of the people alone at their table smiled, laughed, waved ostensibly or talked at the robot in ways similar to behaviors exhibited by people in company. On the whole, people who were alone at their tables were less likely to engage socially with the robot. We suggest that social proof (Cialdini 2007) is missing here. Cialdini proposes that in case of uncertainty, people will look at other people in order to infer how to behave. While people in company thus check with their co-participants what the appropriate way to respond to the robot could be, people who are not in company have no one else to interactively construe and ratify a particular way to categorize the novel experience. Thus, social proof is missing, and consequently many participants decided rather to ignore the robot or to respond to it only with subtle social cues.

Crucially for robot design is that people generally display their lack of willingness to attend to and interact with the robot by withholding social signals. Thus, in order to appropriately negotiate an approach, a robot needs not only be able to read social signals, such as body orientation, gaze direction, waving or speech, but it also needs to have a model of what could be expected if people wanted to interact with it and interpret the absence of such cues as informative regarding people's willingness to engage with the robot. Coordination thus does not only rely on reading social signals, but also on acknowledging their absence.

Another observation concerns the fact that while the audio was most of the time insufficient to tell for certain, it seems that very few participants greeted the robot. In contrast, several thanked the robot after the acknowledgement wiggle. Thus, participants engage only socially in the CASA sense (Nass & Moon 2000) if the robot has displayed contingent social behavior, and some do not at all (e.g. the participant in Fig. 9).

Limitations

In this study, the robot was offering a service to potential participants that most of them were likely to be in use of. This will most certainly have an impact on the transferability of the results of this study to other robots. However, we suspect that the problems regarding the unwillingness to interact with the robot become even more pronounced if the robot is approaching people in its own interest, for instance, to get help, or if people are not just engaged in social conversation but in serious business (cf. Satake et al. 2009, Fischer et al. 2014a).

On the other hand, a study by Lee et al. (2012) on the long-term effects of a snackbot in an office environment shows that while initial excitement may wear off, many people looked forward to the interactions with the robot, arranged to be there when the robot was expected and used the robot's visit as an occasion to socialize. On the whole, the authors find increased social behavior (such as smiling and relationship statements) and a decrease in anti-social behaviors over time. This suggests that the novelty of the robot may not play a decisive role and that instead the kinds of interactions observed in this study are likely to persist also during long term employment of the robot.

As the results by Fischer et al. (2014a) suggest, the use of speech may have had a different effect on the initiation of interactions; their study demonstrates that people respond more readily to a robot when it uses speech. However, given the general noise level in the cafeteria investigated, the employment of state-of-the-art speech technology in such contexts is quite unrealistic; nevertheless, approaching people using speech would certainly have provided those people who did not want to interact with the robot with an affordance of how to signal to the robot that its service is not desired.

Conclusion

The field trial shows that people engaged readily with the robot if they had trash, yet if they hadn't any, they found it difficult to signal this to the robot. Some people even produced trash for the robot even though they had not intended to throw away what they 'fed' to the robot. In general, the robot was understood to 'want' trash, which means that its behavior was understood as an indirect request. We had designed the robot's approaching behavior to negotiate personal space and to produce an acknowledgement when it was used. However, we observed that this kind of negotiation is not relevant if people want the service and have something to throw away. Yet if they don't, then negotiation of space is not as relevant as negotiating whether interaction is desired at all. A similar problem occurs if people are busy and involved in their own activity. Here, the problem is two-fold: on the

one hand a request is a face-threatening act and thus requires interactional labor busy people (or people missing social proof) may not be willing to provide, on the other hand the robot lacks the affordance of rejection: it is not obvious how one can tell the robot to leave the user alone. People's intuitive way to signal that no interaction is desired is by withholding signals, which may be very difficult for a robot to detect.

Design Implications

Carefully approaching people to offer its help seems to be an adequate strategy for a robot if people are willing to interrupt their (social) activity and if they want the service offered. Problems only appear in cases in which people are not willing to socially engage with the robot and/or don't want or need the service. Here, robot designers need to provide for clear communication of which interaction modality will work to make the robot leave, i.e. design for no interaction. Furthermore, robot designers need to equip robots with a model of expected social signals in order to enable the robot to understand the withholding of such signals as a social cue as well.

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